

VoltStream RAG Quality Evaluation Report

Generated on: 2026-06-15 04:37:35 | Predefined QA Suite

Summary Metrics:

Total Questions: 10 Passed: 10 Failed: 0
Pass Rate: 100.0%

Question 1: What is the most cost-effective efficiency upgrade to reduce heating and cooling energy loss?

Answer: Using caulk and weatherstripping around windows and doors is the most cost-effective efficiency upgrade, as up to 30% of heating and cooling energy is lost through small air leaks.

Scores -> Faithfulness: 5/5 | Relevance: 5/5 | Completeness: 5/5 Status: PASS

Question 2: By how much can setting the thermostat to 18 degrees C during winter nights reduce annual heating costs?

Answer: Setting your thermostat to 18 degrees C (64 degrees F) during winter nights or while away can reduce annual heating costs by nearly 10%.

Scores -> Faithfulness: 5/5 | Relevance: 5/5 | Completeness: 5/5 Status: PASS

Question 3: What are the main billing components of an electricity bill?

Answer: Electricity bills consist of fixed connection charges, energy consumption charges (measured in kWh), and demand charges.

Scores -> Faithfulness: 5/5 | Relevance: 5/5 | Completeness: 5/5 Status: PASS

Question 4: When is the best time to run heavy loads to maximize self-consumption with solar offsets?

Answer: To maximize self-consumption with solar offsets, the best time to run heavy loads is when solar generation is highest, which is typically between 11 AM and 2 PM.

Scores -> Faithfulness: 5/5 | Relevance: 5/5 | Completeness: 5/5 Status: PASS

Question 5: What are the typical peak electricity demand hours during summer and winter months?

Answer: Peak electricity demand hours typically occur between 7 PM and 9 PM during summer and winter months.

Scores -> Faithfulness: 5/5 | Relevance: 5/5 | Completeness: 5/5 Status: PASS

Question 6: How can you significantly reduce billing costs related to heavy appliance usage?

Answer: You can significantly reduce billing costs related to heavy appliance usage by shifting their use (e.g., washing machines, dishwashers, and water heaters) to off-peak hours, which are after 10 PM or before 6 AM.

Scores -> Faithfulness: 5/5 | Relevance: 5/5 | Completeness: 5/5 Status: PASS

Question 7: How frequently do smart meters transmit usage data to the utility provider?

Answer: Smart meters transmit usage data to the utility provider every 15 minutes.

Scores -> Faithfulness: 5/5 | Relevance: 5/5 | Completeness: 5/5 Status: PASS

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Question 8: How much of residential energy usage is typically accounted for by phantom or standby load?

Answer: Standby load accounts for up to 10% of residential energy usage.

Scores -> Faithfulness: 5/5 | Relevance: 5/5 | Completeness: 5/5

Status: PASS

Question 9: What is the recommended panel angle for peak annual performance of a residential solar array?

Answer: For peak annual performance of a residential solar array, the recommended panel angle should be equal to the local latitude, generally between 30 degrees and 45 degrees.

Scores -> Faithfulness: 5/5 | Relevance: 5/5 | Completeness: 5/5

Status: PASS

Question 10: How much does cleaning solar panels every 3 months increase overall output efficiency compared to uncleaned panels?

Answer: Cleaning solar panels every 3 months increases overall output efficiency by 15% compared to uncleaned panels.

Scores -> Faithfulness: 5/5 | Relevance: 5/5 | Completeness: 5/5

Status: PASS